



Physiology of the Liver — MCQ Practice

Physiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Approximately what percentage of hepatic blood flow comes from the portal vein?

EASY

- A. 25–30%
- B. 50%
- C. 70–75%
- D. 95%
- E. 10%

2. Which cells are the resident macrophages of the liver, providing immune defense within the sinusoids?

EASY

- A. Hepatocytes
- B. Kupffer cells
- C. Stellate cells
- D. Cholangiocytes
- E. Endothelial cells

3. Which liver function test pattern reflects hepatocellular damage rather than cholestasis?

EASY

- A. ALP and GGT
- B. Albumin only
- C. ALT and AST
- D. Total bilirubin alone
- E. Direct bilirubin alone

4. What is the most common cause of acute liver failure discussed in this lecture?

EASY

- A. Viral hepatitis A
- B. Paracetamol (acetaminophen) overdose
- C. Wilson's disease
- D. Alpha-1 antitrypsin deficiency
- E. Budd-Chiari syndrome

5. A patient with severe liver failure has markedly low albumin and prolonged clotting times but a normal serum immunoglobulin level. How is the normal immunoglobulin level best explained?

MEDIUM

- A. Immunoglobulins are also synthesized by hepatocytes, which are only partially damaged
- B. Immunoglobulins are produced by plasma cells, not hepatocytes, so hepatic synthetic failure does not affect their level
- C. The liver failure is too mild to affect any protein synthesis
- D. Immunoglobulins are stored in the liver, not synthesized there
- E. This patient must not actually have liver failure

6. A patient with liver failure is noted to have a normal Factor VIII level despite low levels of Factors II, VII, IX, and X. What best explains this discrepancy?

MEDIUM

- A. Factor VIII is synthesized by endothelial cells, not hepatocytes, so it is spared in hepatocellular failure
- B. Factor VIII has the longest half-life of all clotting factors
- C. This is a laboratory error, since all clotting factors come from the liver
- D. Factor VIII is vitamin K dependent, so it is regulated separately
- E. Factor VIII is produced in the spleen

7. A patient in a prolonged fast has low blood glucose. Which enzyme limitation explains why the liver, unlike the brain, does not aggressively consume glucose for its own energy even when glucose is scarce?

MEDIUM

- A. The liver lacks glycogen entirely
- B. The liver uses glucokinase rather than hexokinase, so it cannot efficiently trap glucose at low blood concentrations the way brain/muscle can
- C. The liver has no glucose transporters
- D. The liver preferentially burns amino acids at all times
- E. The liver lacks the glycolytic pathway

8. A patient with diabetic ketoacidosis has very high circulating ketone body levels. Why does the liver, despite generating these ketones, not use them as a fuel source itself?

MEDIUM

- A. The liver lacks thiophorase (succinyl-CoA:3-ketoacid CoA transferase), the enzyme required to activate ketone bodies for oxidation
- B. The liver has no mitochondria
- C. Ketone bodies are toxic to hepatocytes specifically
- D. The liver preferentially uses ketones over glucose
- E. This scenario does not occur physiologically

9. A patient has chronic hepatitis B progressing toward cirrhosis. Their serum albumin is falling and ascites has developed. Which liver function is most directly responsible for the ascites via loss of oncotic pressure?

MEDIUM

- A. Bile secretion
- B. Urea cycle detoxification
- C. Albumin synthesis
- D. Kupffer cell phagocytosis
- E. Vitamin D hydroxylation

10. A patient develops confusion and asterix in the setting of advanced cirrhosis. Which liver function, when impaired, is most directly implicated in this presentation?

MEDIUM

- A. Bile acid secretion
- B. The urea cycle, leading to ammonia accumulation
- C. Cholesterol synthesis
- D. Complement production
- E. Vitamin B12 storage

11. A patient's liver panel shows ALP and GGT markedly elevated, out of proportion to only mildly elevated ALT/AST. Which process does this pattern suggest?

MEDIUM

- A. Acute hepatocellular necrosis
- B. Cholestasis (impaired bile flow)
- C. Isolated albumin synthetic failure
- D. Hemolysis
- E. Pure fatty infiltration with no biliary involvement

12. A patient with right heart failure develops hepatic congestion. Which category of liver injury does this best represent?

MEDIUM

- A. Infectious
- B. Toxin/drug-induced
- C. Metabolic/genetic
- D. Vascular
- E. Neoplastic

13. A patient with Budd-Chiari syndrome has thrombosis of which vessel, and what is the direct physiological consequence?

MEDIUM

- A. Portal vein thrombosis, causing pre-hepatic portal hypertension only
- B. Hepatic artery thrombosis, causing ischemic hepatocyte necrosis
- C. Hepatic vein thrombosis, causing outflow obstruction and congestion
- D. Inferior mesenteric vein thrombosis, with no hepatic effect
- E. Splenic vein thrombosis, causing splenomegaly only

14. A neonate has a physiologic reason for some maternal counseling about "liver immune function" that does not apply to adults. What is the basis for this historical teaching point?

MEDIUM

- A. The adult liver actively produces immunoglobulins during infection
- B. The fetal liver performs hematopoiesis, including B-cell precursor development, up to about 6 months gestation — a function lost by adulthood
- C. The neonatal liver contains permanent lymphoid follicles
- D. This is simply incorrect and has no physiological basis
- E. The adult liver stores maternal antibodies

15. A patient with cirrhosis has low albumin, low clotting factors II/VII/IX/X, but normal complement (C3/C4) initially, which later also falls as disease progresses. What does the initial preservation and later fall of complement most likely reflect?

HARD

- A. Complement is never made by the liver, so its later fall must be from an unrelated cause
- B. Complement, like albumin and clotting factors, is hepatically synthesized; its later decline reflects progression of synthetic failure beyond the proteins with shorter half-lives or higher initial reserve
- C. Complement levels are entirely unrelated to hepatic synthetic capacity
- D. The later fall in complement indicates a new infection, not liver disease
- E. Complement is only made by Kupffer cells, unrelated to hepatocyte function

16. A pharmacology student notes that the liver both synthesizes clotting factors and clears activated clotting factors from circulation. In advanced liver failure, which combination of derangements would be expected?

HARD

- A. Only reduced synthesis, with clearance unaffected
- B. Both reduced synthesis of clotting factors AND reduced clearance of activated factors, creating a complex, sometimes paradoxical coagulation picture (bleeding risk from low factors, but also risk of dysregulated clotting)
- C. Only reduced clearance, with synthesis unaffected
- D. Neither synthesis nor clearance is affected in liver failure
- E. Liver failure has no relationship to coagulation at all

17. A researcher compares drug metabolism capacity in a patient with pure hepatocellular disease versus one with pure cholestatic disease at similar severity by lab markers. Which prediction about Phase I/Phase II biotransformation is most defensible?

HARD

- A. Both patients will have identical biotransformation capacity, since biotransformation is unrelated to hepatocyte integrity
- B. The hepatocellular disease patient is more likely to have impaired Phase I/II biotransformation, since these reactions occur within hepatocytes, whereas cholestasis primarily impairs bile excretion rather than the oxidative/conjugative machinery itself
- C. The cholestatic patient will have worse biotransformation capacity in every case
- D. Biotransformation capacity can be inferred directly and precisely from ALP alone
- E. Neither disease pattern has any bearing on drug metabolism

18. A patient with chronic alcohol use progresses through fatty liver, then alcoholic hepatitis, then cirrhosis over years. At the fatty liver stage, LFTs are only mildly deranged. What does this progression best illustrate about the liver's functional reserve?

HARD

- A. The liver has essentially no functional reserve, so any injury immediately causes lab abnormalities
- B. The liver has substantial functional reserve, so early or mild injury (fatty change) may not yet produce marked derangement in standard LFTs, which primarily rise with more advanced hepatocellular injury or cholestasis
- C. LFTs are entirely insensitive to any stage of liver disease
- D. Fatty liver is unrelated to alcohol and this progression is coincidental
- E. Cirrhosis develops independently of the preceding fatty liver and hepatitis stages

19. A patient with hemochromatosis has progressive iron deposition in hepatocytes. Which category of liver damage does this represent, and why might early-stage iron overload be missed by ALT/AST alone?

HARD

- A. Infectious category; ALT/AST are always elevated from the start
- B. Metabolic/genetic category; early iron deposition may cause organelle stress without necessarily producing the degree of hepatocyte necrosis needed to markedly elevate ALT/AST
- C. Vascular category; ALT/AST are irrelevant to any vascular process
- D. Neoplastic category; ALT/AST only rise once metastases are present
- E. Toxin/drug category; hemochromatosis is caused by a hepatotoxic drug

20. A 55-year-old with longstanding untreated chronic HBV infection and cirrhosis develops a new liver mass. Reasoning through the causal chain described in this lecture's categories of liver damage, which sequence best explains the progression from initial infection to this new finding?

HARD

- A. Infectious category (chronic HBV) → cirrhosis → neoplastic category (hepatocellular carcinoma), reflecting the described link between chronic viral hepatitis, cirrhosis, and HCC
- B. Vascular category directly causes HCC without any intermediate steps
- C. Metabolic category alone explains HCC in this patient, independent of the viral infection
- D. This mass is unrelated to the chronic HBV and must be a metastasis from a separate source with no link to the underlying liver disease
- E. Toxin-induced injury is the only pathway that leads to HCC

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	The portal vein supplies 70–75% of hepatic blood flow (nutrient-rich, oxygen-poor).
2	B	EASY	Kupffer cells are the liver-resident macrophages.
3	C	EASY	ALT and AST reflect hepatocellular damage; ALP and GGT reflect cholestasis.
4	B	EASY	Paracetamol overdose is the most common cause of acute liver failure.
5	B	MEDIUM	Unlike albumin and clotting factors, immunoglobulins are made by plasma cells (a classic exam trap), so they remain normal despite hepatocellular synthetic failure.
6	A	MEDIUM	Factor VIII is made by endothelial cells, not hepatocytes, so it is characteristically preserved in hepatocellular liver failure while hepatically-synthesized factors fall.
7	B	MEDIUM	Glucokinase has a much higher K_m than hexokinase, meaning the liver only phosphorylates glucose efficiently when levels are high, sparing glucose for other tissues when it is scarce.
8	A	MEDIUM	The liver lacks thiophorase, so despite producing ketone bodies during fasting/DKA, it cannot use them for its own energy — they are exported for use by other tissues.
9	C	MEDIUM	Hypoalbuminemia from impaired hepatic synthetic function lowers plasma oncotic pressure, contributing to edema and ascites.
10	B	MEDIUM	Hepatic encephalopathy results from ammonia accumulation when the urea cycle can no longer adequately detoxify nitrogenous waste.
11	B	MEDIUM	A disproportionate ALP/GGT rise relative to ALT/AST is the classic cholestatic pattern.
12	D	MEDIUM	Passive congestion from right heart failure is a vascular cause of liver damage, alongside ischemia ("shock liver") and Budd-Chiari syndrome.
13	C	MEDIUM	Budd-Chiari syndrome is hepatic vein thrombosis, causing venous outflow obstruction and hepatic congestion.
14	B	MEDIUM	The fetal liver is a site of hematopoiesis (including B-cell precursors) up to ~6 months gestation, which is the origin of some textbooks' mention of fetal liver "immune functions" — but the adult liver does not make immunoglobulins.
15	B	HARD	The liver synthesizes complement proteins alongside albumin and clotting factors; differential rates of decline across hepatically-synthesized proteins as synthetic failure progresses is consistent with progressive hepatocellular dysfunction, not a separate process.
16	B	HARD	The liver both produces clotting factors and clears activated coagulation intermediates and anticoagulant proteins; failure of both processes explains the complex "rebalanced" but fragile coagulation state seen in cirrhosis.

17	B	HARD	Biotransformation (Phase I oxidation/reduction, Phase II conjugation) is an intrinsic hepatocyte function; hepatocellular injury more directly threatens this machinery, while cholestasis primarily impairs the excretory (bile) pathway rather than the biotransformation reactions themselves — though severe or prolonged cholestasis can secondarily injure hepatocytes too.
18	B	HARD	The liver's large functional reserve means early/mild pathology (steatosis) can exist with near-normal LFTs; measurable derangement typically appears as hepatocellular injury (hepatitis) or synthetic/cholestatic failure (cirrhosis) advances.
19	B	HARD	Hemochromatosis is a metabolic/genetic cause of liver damage; progressive iron deposition can cause subclinical organelle injury before enough hepatocyte necrosis occurs to substantially elevate ALT/AST, which is why screening relies on iron studies (ferritin, transferrin saturation) rather than LFTs alone.
20	A	HARD	The lecture links chronic HBV/HCV infection to cirrhosis, and hepatocellular carcinoma is explicitly linked to hepatitis and cirrhosis — this is the classic causal chain from infectious to neoplastic liver disease.